CIVITAS

Sustainable and smart mobility for all

2030

CIVITAS indicators

Cost per delivery (TRA_FR_EE)

DOMAIN

Transport	Environment	Energy	Society	Economy

TOPIC	**Freight**	
IMPACT	**Economic efficiency of urban deliveries** *Improved economic performance of urban freight distribution through a reduction in the average operational cost required to complete a delivery.*	**TRA_FR**

Category

Key indicator	Supplementary indicator	State indicator

CONTEXT AND RELEVANCE

Urban freight measures frequently aim to improve the efficiency of delivery operations, reduce operational costs for logistics operators, and maintain or improve service levels for businesses and citizens. Actions such as route optimisation, urban consolidation centres, collaborative logistics schemes, digital planning tools, delivery lockers, cargo bike deployment, and improved loading infrastructure can contribute to lower delivery costs.

This indicator measures the average cost incurred by logistics operators to complete a delivery in the pilot area. **It is relevant when the policy action aims to improve the economic efficiency of urban freight operations. A successful action is reflected in a LOWER value of the indicator after the experiment compared to the business-as-usual (BAU) case.**

DESCRIPTION

The indicator assesses the average operational cost associated with completing a delivery. It reflects the efficiency of delivery operations and captures the combined effects of vehicle utilisation, routing efficiency, labour productivity, fleet composition, and operational organisation.

The indicator should be calculated using data collected by surveying delivery operators across a representative sample of delivery tours in the pilot area during a chosen observation period. For each sampled tour, operators should report the operational costs incurred and the number of completed deliveries. The indicator is calculated by dividing the total operational costs reported for the sampled tours by the corresponding total number of completed deliveries.

The indicator is expressed as **EUR/delivery**.

METHOD OF CALCULATION AND INPUTS

The indicator should be calculated **exogenously**. The user computes the indicator value outside the CIVITAS Evaluation tool and enters the final result into the tool.

Method 1	
Survey delivery companies on a sample of delivery tours	Significance: **0.75**

INPUTS

The following information is needed to compute the indicator:

- Total **operational costs** associated with the surveyed delivery tours during the observation period (EUR). This should include labour and vehicle expenses, as well as any other relevant cost relevant to assessing the effects of the pilot measure, such as road tolls, dispatching, administrative costs.
- Total **number of completed deliveries** during the observation period.

EQUATIONS

The equation that should be applied to quantify the indicator is:

$$\boldsymbol{CPD} = \frac{TC}{D}$$

Where:

$\boldsymbol{CPD}$ = Cost per delivery (EUR/delivery)

TC = Total operational delivery costs during the observation period (EUR)

D = Total number of completed deliveries during the observation period

ALTERNATIVE INDICATORS

This indicator measures the average cost associated with completing urban freight deliveries. To assess the operational efficiency factors that influence delivery costs, the CIVITAS Evaluation Framework also provides **TRA_FR_EF1**: Average distance per delivery, **TRA_FR_EF2**: Share of unsuccessful deliveries, **TRA_FR_EF3**: Average time per delivery, and **TRA_FR_EF4** and **TRA_FR_EF5**, which measure the average load factor of delivery vehicles for B2B and B2C deliveries, respectively